



October 30, 2025

Ms. Claire Lowe
Michigan Department of Environment, Great Lakes, and Energy
Water Resources Division
Gaylord District Office
2100 West M-32
Gaylord, MI 49735

**RE: City of Cadillac WWTP (NPDES Permit No. MI0020257)
Revised MAHL Study – Sampling Work Plan – Additional Parameters**

Dear Ms. Lowe:

The City of Cadillac completed a Maximum Allowable Headworks Loading (MAHL) report in May 2024. Per EGLE's direction the City is planning to add phenol, beryllium, benzidine, and bromomethane to its Local Limits. The City has retained Fleis & VandenBrink (F&V) to complete this work for the Cadillac Wastewater Treatment Plant (WWTP).

The City has collected analytical data at the WWTP influent and final effluent over the last two years for phenol, beryllium, benzidine, and bromomethane to complete headworks calculations based on total WWTP removal efficiencies for these parameters. There is limited historic data for these parameters for the permitted industrial users.

Project specific sampling is needed at representative locations within the collection system to differentiate between readily controllable industrial pollutant loadings and corresponding domestic background loadings. Sampling will also be needed at several locations within the WWTP, and the resulting pollutant concentration data will be analyzed to determine removal efficiencies for pollutants of concern (POCs) for which there is no historic data available.

We propose the following Sampling Work Plan in accordance with recommendations from the EPA Local Limits Development Guidance document and 40 CFR Part 136.

Proposed Sampling Plan

MAHL sampling will be completed by City staff and will consist of six 24-hour composite samples at each of the locations listed below.

Four grab samples will be collected in place of each composite sample for parameters for which composite sampling is not appropriate in accordance with 40 CFR Part 136, such as phenol, benzidine and bromomethane.

603 Bay Street, First Floor
Traverse City, MI 49684
P: 231.932.8600
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Location	Parameters	Samples
Representative Domestic #1	All POCs (see list below)	6
Representative Domestic #2	All POCs	6
WWTP Raw Influent	All POCs	6
WWTP Primary Effluent	All POCs	6
WWTP Final Effluent	All POCs	6

Sampling will be scheduled for different days of the week for four weeks (e.g. Monday / Tuesday week 1, Wednesday week 2, Thursday week 3, and Friday / Saturday week 4). The City intends to begin sampling on November 17 (if approved by EGLE). No sampling will be conducted during periods of significant rainfall (>0.5 inches in a 24-hour period) to avoid potential dilution of pollutants.

In accordance with the previous completed MAHL 2024, the background/domestic sampling locations will be the same for the additional parameters. The location for the background / domestic sampling location #1 will be located at Holly Rd and Vine St. The location for the background/domestic sampling location #2 is located at 3rd Avenue.

The City also proposes collecting 2 grab samples at each of the permitted IU locations to check for the presence of the four POCs and gain insight into their concentrations in the discharges.

Location	Parameters	Samples
AAR 6 th Street	All POCs	2
AAR Facility	All POCs	2
AAR Rinse Tank	All POCs	2
Arvco Container Corp.	All POCs	2
AKWEL Cadillac	All POCs	2
Borg Warner of Cadillac	All POCs	2
Cadillac Castings Inc.	All POCs	2
Cadillac Renewable Energy	All POCs	2
Flamm Technologies	All POCs	2
Hutchinson Antivibration	All POCs	2
Michigan Rubber Products	All POCs	2
Piranha Hose Products	All POCs	2

The POCs are listed below with their associated EPA analysis method numbers:

- Phenol – EPA 625.1
- Beryllium – EPA 200.8 Rev. 5.4
- Benzidine – EPA 625.1 SIM
- Bromomethane – EPA 624.1

Please call or email me with any questions or comments as you review the proposed plan.

Sincerely,

FLEIS & VANDENBRINK



Elaine J. Venema, P.E.
Project Manager

cc: Jeff Dietlin, City of Cadillac
Cindy Tomaszewski, City of Cadillac
Patrick Miller, City of Cadillac